

Smart And Secure Mutual Authentication Mechanism For Resource Constrained Iot Devices

*A Report Submitted in the Partial
Fulfillment of the Requirements for the
Completion of the*

eBPL INTERNSHIP

IN

VLSI

Submitted by

D.Varun	23881A0412
E.Pranay Laxman Goud	23881A0413
J.Vamshi	23881A0418

SUPERVISOR

Mr. R.Phani Vidyadhar
Assistant Professor



Department of Electronics and Communication Engineering

June, 2026



Department of Electronics and Communication Engineering

CERTIFICATE

This is to certify that the project titled **Smart And Secure Mutual Authentication Mechanism For Resource Constrained Iot Devices** is carried out by

D.Varun	23881A0412
E.Pranay Laxman Goud	23881A0413
J.Vamshi	23881A0418

in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering** during the year 2025-26.

Signature of the Supervisor
Mr. R.Phani Vidyadhar
Assistant Professor

Signature of the HOD
Dr. S. Rajendar
Professor and Head, ECE

Acknowledgements

The satisfaction that accompanies the successful completion of the task would be put incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

We wish to express our deep sense of gratitude to **Mr. R.Phani Vidyadhar**, Assistant Professor and Project Supervisor, Department of Electronics and Communication Engineering, Vardhaman College of Engineering, for his able guidance and useful suggestions, which helped us in completing the project in time.

We are particularly thankful to **Dr. S. Rajendar**, the Head of the Department, Department of Electronics and Communication Engineering, his guidance, intense support and encouragement, which helped us to mould our project into a successful one.

We show gratitude to our honorable Principal **Dr. J.V.R. Ravindra**, for providing all facilities and support.

We avail this opportunity to express our deep sense of gratitude and heartfelt thanks to **Dr. Teegala Vijender Reddy**, Chairman and **Sri Teegala Upender Reddy**, Secretary of VCE, for providing a congenial atmosphere to complete this project successfully.

We also thank all the staff members of Electronics and Communication Engineering department for their valuable support and generous advice. Finally thanks to all our friends and family members for their continuous support and enthusiastic help.

D.Varun

E.Pranay Laxman Goud

J.Vamshi

Abstract

The Internet of Things (IoT) has enabled billions of interconnected devices to communicate and exchange data, but ensuring security in resource-constrained systems remains a critical challenge. Radio Frequency Identification (RFID) technology plays a vital role in device identification, making secure mutual authentication essential. This project presents a smart and secure mutual authentication mechanism designed specifically for such constrained environments and implemented on an FPGA platform. It enhances an existing protocol by introducing dynamic selection input generation using XOR-based randomness instead of fixed inputs, a 20-bit session key generation module for securing post-authentication communication, and a hardware-based brute-force protection mechanism with a three-attempt lockout feature. The entire system is modeled in Verilog and implemented on a Xilinx Virtex-7 XC7VX485T FPGA using the Vivado Design Suite, comprising encoder modules, a session key generator, and a fault detector. Experimental results demonstrate excellent efficiency, with less than 1% resource utilization, low power consumption of 0.247 W, and strong timing performance. The proposed design effectively resists replay, brute-force, man-in-the-middle, and fault injection attacks, making it a lightweight, secure, and practical solution for RFID-enabled IoT systems.

Keywords: RFID, Lightweight Authentication, IoT Security, FPGA, Verilog HDL, LFSR, Multifunction Circuit, Session Key, Brute-Force Protection, Virtex-7.

Table of Contents

Title	Page No.
Acknowledgements	i
Abstract	ii
List of Figures	vi
Abbreviations	vi
CHAPTER 1 Introduction	1
1.1 Introduction	1
1.2 Background and Motivation	3
1.3 Objectives of the Project Work	5
1.4 Organization of the Report	5
CHAPTER 2 Literature Survey	7
2.1 Introduction	7
2.2 Review of Prior Research	7
2.2.1 Early Lightweight Protocols Based on Bitwise Operations	8
2.2.2 PadGen-Based Protocols	8
2.2.3 Vijaykumar et al. Series of Hardware Protocols	8
2.2.4 Advanced Security Protocols	9
2.2.5 IoT-Specific Authentication Extensions	9
2.3 Identified Research Gaps	9
2.4 Summary	10
CHAPTER 3 Review of Prior Research	12
3.1 Introduction	12
3.2 Multifunction Circuit Architecture	12
3.3 Mutual Authentication Protocol	13
3.4 LFSR for Random Number Generation	15
3.5 Implementation Results of Base Paper	16
3.6 Summary	17
CHAPTER 4 Smart and Secure mutual Authentication mechanism for resource constrained IOT devices	18
4.1 Introduction	18
4.2 System Architecture Overview	19

4.3	Dynamic Selection Input Generation	20
4.4	Session Key Generation Module.....	21
4.5	Fault Injection and Brute-Force Protection	22
4.6	Verilog Implementation Details	23
4.6.1	Multifunction Circuit Module	23
4.6.2	Encoder Module.....	24
4.6.3	Session Key Generator Module.....	24
4.6.4	Top-Level Module.....	24
4.7	Summary.....	25
CHAPTER 5	Results and Discussion	26
5.1	Introduction.....	26
5.2	RTL Simulation Results	27
5.2.1	Waveform Analysis–Successful Authentication	27
5.2.2	Waveform Analysis–Failure and Lockout.....	28
5.2.3	Power Analysis	28
5.3	Timing Analysis	29
5.4	Resource Utilization	30
5.5	Comparison with State-of-the-Art Protocols	30
5.6	Security Analysis.....	32
5.6.1	Replay Attack Resistance.....	32
5.6.2	Brute-Force Attack Resistance	32
5.6.3	Man-in-the-Middle Attack Resistance	33
5.6.4	Session Key Security.....	33
5.6.5	Fault Injection Resistance.....	33
5.7	Summary.....	34
CHAPTER 6	Conclusions and Future Scope	35
6.1	Introduction.....	35
6.2	Summary of Work.....	36
6.3	Key Observations.....	37
6.4	Advantages of the Proposed System.....	38
6.5	Limitations of the Current Work.....	38
6.6	Future Scope.....	39
6.7	Conclusion	39
CHAPTER 7	Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs).....	41
	Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)	41
7.1	Program Specific Outcomes (PSOs).....	42
7.2	Alignment with Sustainable Development Goals (SDGs).....	43

7.3 Overall Impact	43
--------------------------	----

List of Figures

4.1	System Architecture.....	20
5.1	Authentication success case.....	27
5.2	Authentication failure case.....	28
5.3	power report.....	29
5.4	timing analysis.....	29
5.5	Resource utilization report	30

Abbreviations

Abbreviation	Description
CMOS	Complementary Metal Oxide Semiconductor
RFID	Radio Frequency Identification
IoT	Internet of Things
FPGA	Field Programmable Gate Array
ASIC	Application Specific Integrated Circuit
HDL	Hardware Description Language
LFSR	Linear Feedback Shift Register
XOR	Exclusive OR
LUT	Look-Up Table
CLA	Carry Look-Ahead Adder
CLS	Carry Look-Select Adder
EPC	Electronic Product Code
PWD	Password / Secret Key
RNT	Random Number generated by Tag
RNR	Random Number generated by Reader
ENMFRO	Encrypted Message From Reader to Tag
ENMFTO	Encrypted Message From Tag to Reader
BAN	Burrows–Abadi–Needham (Logic)
SPDL	Security Protocol Description Language
WNS	Worst Negative Slack
TNS	Total Negative Slack
PDP	Power Delay Product
SK	Session Key

CHAPTER 1

Introduction

1.1 Introduction

The Internet of Things (IoT) has emerged as one of the most important technological paradigms of the twenty-first century, enabling seamless communication between physical objects, sensors, and computing devices through the Internet. IoT systems are widely used in applications such as smart healthcare, industrial automation, smart cities, agriculture, and supply chain management. With the rapid growth of digital technologies, the number of connected IoT devices has increased significantly, with projections estimating tens of billions of devices in the near future. Among the enabling technologies, Radio Frequency Identification (RFID) plays a key role in providing identification and communication capabilities for many IoT applications.

An RFID system typically consists of three main components: a tag, a reader, and a backend server. The RFID tag is attached to an object and contains a unique identifier, while the reader communicates with the tag using radio frequency signals. The backend server processes the data received from the reader and performs necessary operations such as authentication and data storage. This architecture enables contactless communication and makes RFID suitable for applications such as inventory management, access control, asset tracking, and electronic toll collection. However, the wireless communication channel between the tag and reader is inherently insecure, making the system vulnerable to several types of attacks.

Security is a major concern in RFID-based IoT systems due to the open nature of wireless communication. Attackers can easily intercept communication and perform attacks such as eavesdropping, replay attacks, man-in-the-middle attacks, and tag cloning. These attacks can compromise the confidentiality, integrity, and authenticity of the system. For example, in a replay attack, an

adversary can capture a valid communication session and reuse it later to gain unauthorized access. Similarly, cloning attacks allow attackers to duplicate a legitimate tag, leading to serious security risks. Therefore, it is essential to design secure authentication mechanisms to protect RFID systems from such threats.

Mutual authentication is a fundamental requirement for secure RFID communication. In this approach, both the tag and the reader verify each other's identity before exchanging any sensitive information. This prevents unauthorized devices from accessing the system and ensures secure communication. However, implementing strong authentication in RFID systems is challenging due to the limited resources of passive RFID tags. These tags have very low power consumption, limited memory, and minimal computational capability, as they rely on energy from the reader signal.

Because of these constraints, traditional cryptographic techniques such as public-key cryptography and complex symmetric encryption algorithms cannot be directly used in RFID systems. These methods require high computational power and large hardware resources, making them unsuitable for low-cost tags. To overcome this issue, researchers have developed lightweight authentication protocols that use simple operations such as XOR, bitwise rotations, modular arithmetic, and Linear Feedback Shift Registers (LFSRs). These techniques provide a balance between security and efficiency, making them suitable for resource-constrained environments.

Over time, several lightweight authentication protocols have been proposed to improve both security and performance. Early protocols focused mainly on reducing computational complexity but were later found to be vulnerable to various attacks. Recent approaches have emphasized hardware-based implementations, where protocols are designed and tested using FPGA platforms to evaluate parameters such as area, power consumption, and delay. This shift toward hardware design has enabled more efficient and practical solutions for real-world IoT systems.

One of the recent advancements in this area is the multifunction circuit-based mutual authentication protocol proposed by Vijaykumar et al. (2024).

This protocol introduces a multifunction circuit capable of generating multiple outputs based on selection inputs, thereby improving security without significantly increasing hardware complexity. The design achieves a good balance between low power consumption, reduced area, and enhanced functionality, making it suitable for IoT applications.

However, despite its advantages, the existing protocol has certain limitations. The selection inputs used in the multifunction circuit are fixed in the hardware implementation, which reduces randomness and may lead to potential vulnerabilities. Additionally, the protocol does not generate a session key after successful authentication, leaving subsequent communication unprotected. It also lacks mechanisms to prevent brute-force attacks and does not address fault injection attacks.

To address these challenges, this project enhances the existing protocol by introducing dynamic selection inputs, session key generation, and brute-force attack protection. These improvements aim to increase security while maintaining low hardware complexity. The proposed design is implemented on a Xilinx Virtex-7 FPGA platform, providing better performance and more realistic evaluation compared to older devices. Overall, this work focuses on achieving a secure and efficient authentication mechanism suitable for modern RFID-based IoT systems.

1.2 Background and Motivation

The foundation of this project lies in the observation that existing lightweight RFID authentication protocols, while effective at reducing hardware complexity, often lack critical security features needed for robust IoT deployments. A survey of the literature reveals several recurring limitations.

Early protocols such as LMAP, M2AP, and EMAP proposed by Peris-Lopez et al. relied on simple bitwise operations but were later demonstrated to be vulnerable to de-synchronization and traceability attacks. Konidala et al. introduced the PadGen concept using two passwords and two random numbers to generate cover-coded passwords, but this was a software-only implementation with no hardware realization. Huang et al. subsequently implemented

hardware versions using the PadGen scheme, demonstrating improved area and power efficiency but with moderate security levels. Vijaykumar et al. made significant contributions through truncated multiplier-based and modulo adder-based protocols, both implemented on Altera Cyclone II FPGAs, with formal security proofs using BAN logic.

The most recent work by Vijaykumar et al. (2024), which forms the base paper for this project, introduced a novel multifunction circuit based on a Quad Field Carry Look-Ahead / Carry Look-Select (CLA/CLS) architecture. This circuit produces six different output functions based on three selection inputs (SL1, SL2, ADD/SUB), dramatically increasing the output space and thereby improving resistance to known-plaintext attacks. The protocol was implemented on an Altera Cyclone II FPGA, achieving 66 logic elements, 4.36 mW dynamic power, 10.04 ns delay, and 60.6 pJ PDP — excellent metrics for a lightweight protocol.

However, this base protocol exhibits several security limitations that motivate the enhancements proposed in this project. First, the selection inputs SL1 and SL2 are fixed by the designer at synthesis time, meaning they remain constant across all authentication sessions. This determinism weakens the security of the multifunction circuit, as an adversary who learns the selection inputs can predict the output function regardless of the random numbers used. Second, the protocol provides no mechanism for securing post-authentication data exchange — once mutual authentication succeeds, all subsequent communications proceed without a session-specific encryption key. Third, there is no hardware protection against brute-force attacks; a malicious entity could attempt repeated authentication with guessed passwords without any lockout mechanism.

These gaps provide the primary motivation for the enhanced protocol proposed in this work. Additionally, the advancement to a Xilinx Virtex-7 platform allows us to leverage a more modern and capable FPGA architecture while still demonstrating the lightweight nature of the protocol.

1.3 Objectives of the Project Work

The primary objective of this project is to design, implement, and verify a secure and resource-efficient mutual authentication protocol for RFID-enabled IoT devices that addresses the security gaps identified in existing work. The specific objectives are:

1. To Develop the multifunction circuit-based mutual authentication protocol from Vijaykumar et al. (2024) in Verilog HDL and confirm its functional correctness through RTL simulation.
2. To Introduce a dynamic selection input generation mechanism that derives SL1 and SL2 from XOR operations on bits of the random numbers RNT and RNR, ensuring that the selection inputs vary with every authentication session.
3. To Implement a module that produces a unique 20-bit session key upon successful mutual authentication to enable secure post-authentication data exchange within the system.
4. To Design and implement a fault injection detection and brute-force protection module that counts failed authentication attempts and enforces a lockout after three consecutive failures.
5. To Deploy the design on a Xilinx Virtex-7 XC7VX485T FPGA using Vivado, analyze resource utilization, power consumption, and timing performance, and perform a comprehensive comparison with existing state-of-the-art protocols while verifying security against replay, brute-force, man-in-the-middle, and fault injection attacks.

1.4 Organization of the Report

This report is organized into several chapters to clearly present the background, methodology, implementation, and results of the proposed work.

Chapter 2: Literature Review Deploy the design on a Xilinx Virtex-7 XC7VX485T FPGA using Vivado, analyze resource utilization, power consumption, and timing performance, and perform a comprehensive comparison with existing state-of-the-art protocols while verifying security against replay, brute-force, man-in-the-middle, and fault injection attacks.

Chapter 3: Technical Background of RFID and IoT Authentication Systems describes the base paper protocol in detail, covering the multifunction circuit architecture, the mutual authentication flow, encoder design, and the implementation results reported in the original paper.

Chapter 4: Proposed Lightweight Hardware Authentication Protocol presents the proposed system design, including the architectural enhancements, Verilog module descriptions, and the complete implementation details for all four modules of the proposed system.

Chapter 5: Results and Discussion presents the results of RTL simulation, power analysis, timing analysis, and resource utilization from the Vivado implementation. A detailed comparison with state-of-the-art protocols is also provided.

Chapter 6: Conclusions and Recommendations concludes the report by summarizing the key contributions and suggesting directions for future research.

CHAPTER 2

Literature Survey

2.1 Introduction

The problem of secure authentication in RFID systems has been a major research focus for over two decades [4], [5]. The main challenge lies in providing strong security while respecting the tight resource limits of passive RFID tags. These tags usually have only a few hundred logic gates, extremely low power budgets in the microwatt range, and no persistent state between sessions [5]. Designing lightweight mutual authentication protocols that work under these constraints is therefore critical.

Researchers have explored a variety of approaches to achieve this balance. Protocols are often categorized by their underlying computational methods, such as hash-based, XOR-based, or symmetric-key schemes [8]. Implementation platforms also vary, ranging from simple low-cost tags to FPGA-based experimental systems [1]. Security goals typically include resistance to replay attacks, brute-force attacks, and man-in-the-middle attacks [4]. Efficiency metrics like area utilization, power consumption, and latency are central to protocol evaluation [1].

2.2 Review of Prior Research

Several researchers have proposed authentication protocols to improve the security of RFID-based IoT systems [1], [4]. Early lightweight authentication protocols such as LMAP and EMAP used simple bitwise operations to reduce computational complexity for low-cost RFID tags. Although these protocols were efficient, later studies found that they were vulnerable to various security attacks [4].

2.2.1 Early Lightweight Protocols Based on Bitwise Operations

The EPC Global Class-1 Generation-2 (C1G2) standard employs a simple one-way authentication mechanism using a 16-bit pseudorandom number generated by an LFSR [4], [3]. While this provides a basic level of tag identification, the one-way nature of authentication leaves the system vulnerable to rogue reader attacks and tag cloning [4].

Peris-Lopez et al. proposed ultra-lightweight protocols such as EMAP, LMAP, and M2AP using simple logical operations. However, these protocols were later shown to be vulnerable to de-synchronization and full disclosure attacks [4].

Chien proposed the SASI protocol using XOR, rotation, and modular addition, but it was later broken through algebraic and statistical attacks [4]. These early results highlighted the need for rigorous security validation in lightweight protocol design.

2.2.2 PadGen-Based Protocols

Konidala et al. introduced the PadGen concept, generating encrypted passwords using XOR operations and random numbers. While it improved security, hardware implications were not addressed.

Huang et al. implemented the PadGen protocol on FPGA platforms, demonstrating feasibility but with relatively high area and power consumption [1]. Their designs consumed significant logic elements and power, making them less suitable for constrained IoT devices [1].

2.2.3 Vijaykumar et al. Series of Hardware Protocols

Vijaykumar et al. proposed several hardware-efficient authentication protocols targeting reduced area and power consumption [1].

The truncated multiplier-based approach achieved low area and reasonable randomness for security [1]. A later protocol using a modulo adder architecture further improved efficiency and was formally verified using BAN logic [1].

The most recent work introduced a multifunction circuit capable of generating multiple outputs dynamically, improving security by varying computation per session [1], [7].

2.2.4 Advanced Security Protocols

For higher security, hash-function-based and block cipher-based protocols have been proposed. Lightweight block ciphers improve resistance against sophisticated attacks but require more resources [2], [5].

Physically Unclonable Function (PUF)-based authentication schemes provide strong resistance to cloning attacks by leveraging hardware uniqueness, though at increased hardware cost [5].

2.2.5 IoT-Specific Authentication Extensions

Lightweight authentication has also been extended to broader IoT applications such as drones and integrated networks, using hash-based techniques and formal verification methods like BAN logic [8]. These works demonstrate the adaptability of authentication protocols across IoT domains.

2.3 Identified Research Gaps

A critical analysis of the literature reveals several important research gaps that are addressed by the proposed work:

- **Static Selection Inputs:** In the base protocol (2024), the selection inputs SL1, SL2, and ADD/SUB to the multifunction circuit are fixed at design time by the designer. While the base paper mentions that SL1 and SL2 are derived from random number bits, the actual Verilog implementation and FPGA realization uses fixed values. This determinism creates a vulnerability: an adversary who intercepts multiple authentication sessions can, in principle, determine the fixed selection inputs and simplify the analysis of the encryption function.
- **Absence of Session Key:** The existing base protocol and all prior protocols in the Vijaykumar series do not generate a session key after successful

authentication. This means all post-authentication data exchange (sensor readings, actuator commands) is transmitted without session-specific encryption, leaving the channel vulnerable to eavesdropping even after authentication succeeds.

- **No Brute-Force Protection:** None of the reviewed hardware implementations include a mechanism to detect and respond to repeated failed authentication attempts. An adversary with physical or wireless access to the RF channel can attempt authentication with different guessed passwords indefinitely, with no system response to limit the attack.
- **Platform Limitations:** The majority of hardware implementations target Altera Cyclone II (EP2C35F672C6), a relatively dated FPGA from 2006. Modern IoT security implementations benefit from advanced FPGA platforms that offer better routing resources, faster logic cells, and more sophisticated I/O. A re-evaluation on modern platforms such as Xilinx Virtex-7 provides more relevant benchmarks for contemporary applications.
- **Fault Injection Resilience:** None of the reviewed lightweight protocols address resistance to fault injection attacks, where an adversary deliberately corrupts the computation through power glitching, clock manipulation, or electromagnetic injection. A hardware counter that detects abnormal signal patterns provides a basic but effective first line of defense.

2.4 Summary

This chapter has reviewed the evolution of RFID mutual authentication protocols, tracing the progression from early software-based designs using simple bitwise operations to increasingly sophisticated hardware implementations. Over time, protocols have consistently aimed to reduce area, power consumption, and delay while maintaining or enhancing security. Formal verification has become a standard practice to ensure correctness and robustness. The use of multifunction circuits and lightweight computations has enabled efficient

implementation on resource-constrained devices. Researchers have also focused on balancing security features with the strict limitations of passive RFID tags. This historical perspective sets the foundation for understanding current state-of-the-art approaches.

The base paper by Vijaykumar et al. (2024) represents the current benchmark for lightweight multifunction-circuit-based mutual authentication protocols. However, several research gaps remain, including the use of static selection inputs, absence of session key generation, lack of brute-force protection, and limited resilience to fault injection attacks. These gaps directly motivate the enhancements proposed in this project. By introducing dynamic selection inputs, a session key generator, and hardware-based security mechanisms, the project addresses these limitations. The architectural improvements ensure stronger security without compromising resource efficiency. This work therefore extends the state-of-the-art design to meet modern IoT and RFID security requirements.

CHAPTER 3

Review of Prior Research

3.1 Introduction

This chapter presents the technical background related to RFID-based IoT systems and authentication mechanisms. It explains the basic concepts and technologies used in prior research, including RFID system architecture, authentication protocols, and lightweight security techniques. Understanding these concepts helps in analyzing existing methods and provides the foundation for the proposed lightweight hardware authentication protocol.

In addition to the hardware design, the paper introduces a mutual authentication protocol between the RFID tag, reader, and backend server. The protocol makes use of random numbers generated through Linear Feedback Shift Registers (LFSR) to ensure freshness of communication and prevent security attacks such as replay attacks, tag cloning, and man-in-the-middle attacks. Furthermore, the proposed system is implemented on FPGA platforms to evaluate its performance in terms of power consumption, hardware utilization, and delay. The results demonstrate that the proposed design outperforms existing authentication protocols, making it a reliable solution for secure and efficient IoT communication systems.

3.2 Multifunction Circuit Architecture

The core contribution of the base paper is the design of a novel multifunction circuit that produces six distinct output functions from the same pair of 16-bit inputs, controlled by three selection inputs. This circuit is built around a Quad Field CLA/CLS (Carry Look-Ahead / Carry Look-Select) architecture that processes two 16-bit inputs X and Y along with three control signals: $SL1$, $SL2$, and ADD/SUB .

The proposed circuit processes two 16-bit inputs X and Y with the three

selection inputs SL1, SL2 and ADD/SUB and produces an output Z of 20-bit length. The architecture uses a 4×4-bit dual-field parallel CLA adder as its computational core, which provides fast carry propagation through the carry look-ahead principle rather than the slower ripple-carry scheme. The selection inputs control both the routing of intermediate results and the operational mode of the carry generation logic.

The six output functions produced by the multifunction circuit correspond to different combinations of the selection inputs. When SL1=0 and SL2=0, the circuit performs addition of X and Y. When SL1=1 and SL2=0, subtraction is performed. When SL2=1, the circuit routes through the carry look-select path, producing different combinations of partial sums. The ADD/SUB signal further modifies the carry injection, enabling six distinct computational modes from the same hardware.

The critical security insight is that the selection inputs are derived from bits of the random numbers exchanged during authentication. Since the random numbers change with every authentication session (generated by LFSRs initialized with unique seeds), the selection inputs also change, meaning the computational function applied to the password changes each session. This makes it extremely difficult for an adversary to perform known-plaintext analysis since the function itself is unknown and session-dependent.

The multifunction circuit operates with simple logic gates — AND, OR, XOR, and multiplexers — and employs parallel 4-bit adder stages for fast computation. This keeps the gate count low while delivering a large output function space. The circuit produces a 20-bit output from two 16-bit inputs, providing a 4-bit expansion that contributes additional security margin

3.3 Mutual Authentication Protocol

The base paper's mutual authentication protocol follows a standard three-party RFID architecture involving the Tag, Reader, and Backend Server. The protocol operates in two phases — Reader Authentication and Tag Authentication — as summarized below:

1. The Reader broadcasts a request signal [ReqR] over the RF channel to

energize nearby tags.

2. Any tag within range becomes energized and generates a 64-bit random number RNT using its LFSR. The tag transmits its Electronic Product Code (EPC) and RNT to the Reader.
3. The Reader forwards the received EPC and RNT to the Backend Server through the secure wired channel.
4. The Server retrieves the tag's secret password PWD from its database using the EPC. It generates a new 64-bit random number RNR and computes the encrypted password ENMFRO using the multifunction-based encoder: $ENMFRO = ENCODER(PWD, RNT, SL1, SL2)$.
5. The Server sends ENMFRO, EPC, and RNR back to the Reader, which forwards them to the Tag over the RF channel.
6. The Tag independently computes $IENMFRO = ENCODER(PWD, RNT, SL1, SL2)$ using its own copy of PWD. If $IENMFRO == ENMFRO$, the Reader is authenticated. If not, the protocol terminates.
7. The Tag then computes $ENMFTO = ENCODER(PWD, RNR, SL1, SL2)$ and sends it with the EPC to the Reader, which forwards it to the Server.
8. The Server computes $IENMFTO = ENCODER(PWD, RNR, SL1, SL2)$ and verifies: if $IENMFTO == ENMFTO$, the Tag is authenticated. Mutual authentication is complete.

The encoder generates a 20-bit encrypted password from a 32-bit secret password (PWD) and a 64-bit random number using XOR operations and the multifunction circuit. The equations defining the encoder operations are:

$$IE1[31:0] = PWD[31:0] \text{ xor } RN[31:0]$$

$$IE2[31:0] = PWD[31:0] \text{ xor } RN[63:32]$$

$$IE3[15:0] = IE1[31:16] \text{ xor } IE2[15:0]$$

$$IE4[15:0] = IE2[31:16] \text{ xor } IE1[15:0]$$

$IE5[15:0] = PWD[31:16] \text{ xor } IE3[15:0]$

$IE6[15:0] = PWD[15:0] \text{ xor } IE4[15:0]$

EN $PWD[19:0] = \text{MULTIFUN_CKT}(IE5, IE6, SL1, SL2, \text{ADD/SUB})$

This multi-stage XOR cascade ensures that both halves of the 64-bit random number influence the final inputs to the multifunction circuit. The password is XORed at multiple levels, creating a complex dependency between PWD, RN, and the final encrypted output.

3.4 LFSR for Random Number Generation

The base paper employs Fibonacci Linear Feedback Shift Registers (LFSRs) to generate the pseudo-random numbers RNT (Tag Random Number) and RNR (Reader Random Number), which are essential for ensuring secure and dynamic authentication in IoT-based RFID systems. LFSRs are highly suitable for lightweight hardware environments because they are built using only D flip-flops and XOR gates, which are among the simplest and most power-efficient digital components. This makes them ideal for resource-constrained devices, where minimizing area, power consumption, and design complexity is critical.

In this design, a 64-bit LFSR is used, meaning it consists of 64 flip-flops connected in a shift register configuration. The randomness of the output sequence depends on the choice of the primitive polynomial, which determines the feedback connections. When a proper primitive polynomial is selected, the LFSR produces a maximum-length sequence (MLS), cycling through all possible non-zero states before repeating. For a 64-bit LFSR, this results in a sequence length of $2^{64}-1$, which is extremely large. This period ensures that the generated random numbers do not repeat within any practical usage timeframe, thereby enhancing security and unpredictability.

The implementation follows the Fibonacci configuration, where the feedback is applied externally through XOR logic. Specifically, the feedback bit is generated by XORing the outputs of selected tap positions—in this case, bit positions 63 and 62—and feeding the result back into the input of the shift register. On every clock cycle, the register shifts its contents, and the

new bit enters based on this feedback logic. This continuous shifting and feedback mechanism produces a sequence that appears random, although it is deterministically generated.

Overall, the use of a 64-bit Fibonacci LFSR with a carefully chosen primitive polynomial provides an efficient balance between hardware simplicity, low power consumption, and high-quality pseudo-randomness, making it a strong choice for secure authentication in lightweight IoT systems.

3.5 Implementation Results of Base Paper

The base paper implemented the protocol on an Altera DE2 Cyclone II FPGA (EP2C35F672C6) using Altera Quartus II software. The key metrics reported are:

Table 3.1: Base Paper Performance Metrics

Metric	Base Paper [2024]
Logic Elements	66
Dynamic Power (mW)	4.36
Clock-to-Output Delay (ns)	10.04
PDP (pJ)	60.6
Fault Detection	No
Session Key	No

The base paper also reports ASIC implementation results in both 180 nm and 90 nm Cadence RTL compiler technologies, occupying only 10,122 μm^2 at 180 nm (compared to 16,943 μm^2 for Vijaykumar [14]) and 3,495 μm^2 at 90 nm. the Dynamic power at 180 nm was 906 μW and at 90 nm was 216 μW — extremely low for a security protocol.

The formal security of the protocol was verified using BAN logic, with all security goals (mutual authentication between tag and reader) proven to be satisfied. Additionally, the Scyther automatic verification tool confirmed the absence of known protocol attacks within the model. Informal security analysis confirmed resistance to replay attacks, forward secrecy violation, brute-force attacks, and man-in-the-middle attacks.

3.6 Summary

The base paper presents a well-optimized lightweight mutual authentication protocol that achieves state-of-the-art performance in terms of area, power, and formal security verification. Its multifunction circuit architecture is a clever solution to the problem of maximizing the output function space with minimal hardware. However, as identified in Chapter 2, the fixed selection inputs, lack of session key generation, and absence of brute-force protection limit its applicability in real-world adversarial IoT deployments. These limitations are systematically addressed in the proposed work described in Chapter 4.

In addition, the protocol relies heavily on static operational patterns within the multifunction circuit, which may reduce randomness over multiple authentication sessions and make the system more predictable under advanced attack models. Although the use of LFSR-based random number generation improves efficiency, it may not provide sufficient cryptographic strength against sophisticated attackers when used without additional security layers. Furthermore, the absence of dynamic parameter updates restricts the protocol's ability to adapt to evolving security threats.

CHAPTER 4

Smart and Secure mutual Authentication mechanism for resource constrained IOT devices

4.1 Introduction

This chapter presents the complete architectural design, module descriptions, and Verilog implementation of the proposed enhanced mutual authentication system. The proposed system builds upon the multifunction circuit-based encoder architecture of the base paper and introduces three key enhancements: dynamic selection input generation, session key generation after successful authentication, and hardware-based brute-force and fault injection protection. The complete design is implemented as a hierarchy of Verilog modules under a top-level entity called system top.

The architecture is designed in a modular and scalable manner, where each functional block such as the random number generator, multifunction circuit, control logic, and authentication unit is implemented as an independent Verilog module. This modular approach improves readability, reusability, and ease of debugging during hardware implementation. The interaction between modules is controlled through well-defined input-output interfaces, ensuring synchronized data flow across the system.

The dynamic selection input generation mechanism enhances security by varying the operational behavior of the multifunction circuit in each authentication session, thereby increasing unpredictability. The session key generation module ensures that a unique key is derived after every successful authentication, which can be used for securing subsequent communications. Additionally, the inclusion of hardware-based protection mechanisms such as brute-force detection counters and fault monitoring logic improves the system's robustness against unauthorized access and physical attacks.

4.2 System Architecture Overview

The proposed system consists of four primary functional blocks, as illustrated in the block diagram:

1. **Tag Encoder (tag_enc):** Implements the multifunction-based encoder for computing the encrypted password from the tag's perspective.
2. **Server Encoder (srv_enc):** Implements the same encoder architecture from the server's perspective, generating the expected encrypted password for comparison.
3. **Session Key Generator (skg):** Computes a unique 20-bit session key upon successful mutual authentication, using the same multifunction circuit with a different combination of inputs derived from both RNT and RNR.
4. **Fault Detector (fd):** Monitors authentication outcomes and maintains a 3-bit counter of consecutive failures, asserting AUTH LOCKOUT when the count reaches 3.

The top-level module (system top) coordinates these four blocks and implements the authentication logic, comparing the outputs of the tag encoder and server encoder to determine authentication success or failure. The system takes the following primary inputs: clk (clock), reset (asynchronous reset), RNT IN[63:0] (tag random number), RNR IN[63:0] (reader/server random number), inject fail (test input for fault injection simulation), and PWD[31:0] (32-bit secret password).

The primary outputs are: AUTH_SUCCESS (authentication passed), AUTH_FAIL (authentication failed), AUTH_LOCKOUT (system locked due to excessive failures), SK_VALID (session key valid), SESSION KEY[19:0] (the generated session key), ENMFRO[19:0] and ENMFTO[19:0] (encrypted messages), RNT[63:0] and RNR[63:0] (internal registered copies of inputs), and fail count[2:0] (current failure count).

any particular guess is of no value for future sessions.

The dynamic sl gen module is combinational it receives RNT and RNR as inputs and produces SL1, SL2, and ADD SUB as registered outputs on the clock edge. By registering the outputs, the selection inputs are synchronized to the same clock domain as the rest of the design, preventing metastability issues.

4.4 Session Key Generation Module

The session key generation module (skg or session key gen) is a novel addition that produces a 20-bit session key upon successful mutual authentication. The session key is intended to be used for encrypting post-authentication data exchange between the tag and the reader, ensuring that even if an eavesdropper captures the authentication exchange, they cannot decrypt subsequent communications without knowledge of the session key.

The session key is generated using the same multifunction circuit architecture as the encoders, but with a different input derivation:

$$sk.x[15:0] = RNT[15:0] \oplus RNR[15:0]$$

$$sk.y[15:0] = RNT[31:16] \oplus RNR[31:16]$$

$$SESSION_KEY[19:0] = MULTIFUN_CKT(sk.x, sk.y, SL1_0, SL2_0, ADD_SUB_0)$$

Where:

$$SL1_0 = RNT[32] \oplus RNR[32]$$

$$SL2_0 = RNT[33] \oplus RNR[33]$$

$$ADD_SUB_0 = RNT[34] \oplus RNR[34]$$

The session key is registered and the SK_VALID flag is asserted only when AUTH SUCCESS is high. The key is cleared (reset to zero) on any reset event or lockout, ensuring that a failed authentication attempt does not leave a stale session key accessible to an attacker. The generation of the session key using XOR combinations of both RNT and RNR ensures that both parties (tag and server) can independently compute the same key — the tag has access to both RNT (which it generated) and RNR (received from the reader), and the server similarly has access to both. An eavesdropper who intercepts the RF channel observes only the encrypted passwords ENMFRO and ENMFTO,

not the random numbers directly (they are transmitted in the clear in this protocol), but the relationship between random numbers and session key still requires knowledge of the exact selection bits to reconstruct.

4.5 Fault Injection and Brute-Force Protection

The fault detector (fd) module provides hardware-level protection against two related attacks: brute-force password guessing and fault injection. The module maintains a 3-bit failure counter (fail_count) that increments on each AUTH FAIL event. When fail count reaches 3 (binary 011), the AUTH LOCKOUT signal is asserted and the system refuses further authentication attempts until a reset is applied.

The brute-force protection mechanism works as follows. Each time an authentication attempt fails (AUTH_FAIL is asserted), the fault detector increments fail count. The progression of fail count is visible in the RTL simulation waveforms: after the first failure, fail_count = 1; after the second, fail_count = 2; after the third, fail count = 3 and AUTH LOCKOUT is set to '1'. Once locked out, even a valid authentication attempt will be blocked until the system is reset.

Fault injection protection is demonstrated through the inject fail input signal. In a real deployment, this would be connected to sensors that detect anomalous power supply variations, unusual clock characteristics, or other environmental signatures of fault injection attacks. In the simulation environment, inject fail serves as a test stimulus to verify that the fault detection logic correctly identifies injected faults and triggers the lockout mechanism.

The fault detector uses a read-only memory (ROM) lookup implemented as a case statement to determine AUTH LOCKOUT based on fail_count. This is more robust than a simple comparison because it can be programmed with different threshold behaviors without changing the logic structure.

4.6 Verilog Implementation Details

4.6.1 Multifunction Circuit Module

The multifunction_circuit module is the computational core used by all three encoder instances (tag_enc, srv_enc, and the session key generator within skg). It takes two 16-bit inputs X and Y, three selection signals (SL1, SL2, ADD_SUB), and produces a 20-bit output Z.

The internal implementation uses a 4×4-bit parallel CLA structure. The carry generate (G) and carry propagate (P) signals are computed for each 4-bit group. The carry look-ahead logic computes all carry signals in parallel based on G and P, eliminating the serial ripple-carry delay. The sum outputs are then selected based on SL1 and SL2 using multiplexers, and the ADD_SUB signal controls the complement of Y for subtraction. The key structural elements are:

- Four parallel 4-bit adder/subtractor units
- Carry look-ahead logic generating C4, C8, C12, C16
- Carry look-select multiplexers (SL1 selects between ripple-carry and CLA paths)
- Output multiplexer (SL2 selects between arithmetic and logic operations)
- 20-bit output formed by carry_out[3:0], sum[15:0]

Table 4.1: Functional Operation Description

SL1	SL2	ADD_SUB	Output Function	Operation Description
0	0	0	$X + Y$	Standard 20-bit addition
0	0	1	$X - Y$	20-bit subtraction
0	1	0	$X \text{ XOR } Y$ (20-bit)	Bitwise OR after zero-extension
0	1	1	$X \text{ XOR } Y$ (20-bit)	Bitwise XOR after zero-extension
1	0	0	$(X + Y)$ via CLS path	Carry Look-Select addition
1	0	1	$(X - Y)$ via CLS path	Carry Look-Select subtraction

4.6.2 Encoder Module

The encoder module (referenced as both `tag_enc` and `srv_enc`, both instantiating the same encoder module) implements the six-stage XOR cascade described in Section 3.3 followed by the multifunction circuit. The module takes `PWD[31:0]`, `RN[63:0]`, `SL1`, `SL2`, `ADD_SUB` as inputs and produces `EN_PWD[19:0]` as output. A 32-bit constant parameter `V` (default `0xA1B2C3D4`) provides an additional XOR mask that can be used to differentiate the tag encoder from the server encoder at instantiation, though in the current implementation both use the same `PWD`.

The encoder is purely combinational—it contains no registers. All computation is done in a single clock cycle, keeping the latency minimal. The combinational path depth is dominated by the multifunction circuit (approximately 6-8 gate levels through the CLA chain), which explains the excellent timing results achieved.

4.6.3 Session Key Generator Module

The `session_key_gen` module wraps the multifunction circuit with additional logic for session key derivation and validity signaling. It contains registered outputs (`SESSION KEY reg`, `SK VALID reg`) that are updated on the rising edge of `clk`. The asynchronous reset (active low in the base design) clears all registers.

When `AUTH_SUCCESS` is asserted, the `SESSION KEY` register is updated with the computed session key value on the next clock edge. The `SK VALID` flag is set simultaneously. When `AUTH_LOCKOUT` is detected or reset is applied, both `SESSION KEY` and `SK VALID` are cleared.

4.6.4 Top-Level Module

The `system_top` module integrates all four submodules and implements the authentication comparison logic. The comparison is done using an equality comparator (`RTL_EQO` in the RTL schematic) between the outputs of `tag_enc` and `srv_enc`. When the outputs match and neither `AUTH_LOCKOUT` nor

inject fail is active, AUTH SUCCESS reg is set to '1' on the next clock edge.

The complete top-level port list is:

Inputs: clk, reset, inject fail, RNT_IN[63:0], RNR_IN[63:0]

Outputs: AUTH SUCCESS, AUTH FAIL, AUTH LOCKOUT, SK_VALID, SESSION_KEY[19:0], ENMFRO[19:0], ENMFTO[19:0], RNT[63:0], RNR[63:0], fail_count[2:0]

The module registers RNT_IN and RNR_IN internally to provide stable values to all submodules throughout the authentication computation. The registered RNT and RNR are also exposed as outputs for testbench monitoring.

4.7 Summary

This chapter presents the complete architectural design and Verilog implementation of the proposed enhanced mutual authentication system. The four modules—tag_enc, srv_enc, skg, and fd—together implement a secure, lightweight mutual authentication protocol with dynamic selection inputs, session key generation, and brute-force protection. The design leverages the proven multifunction circuit architecture from the base paper while addressing all three identified security gaps. The complete system is described in synthesizable Verilog HDL and was successfully synthesized and implemented on a Xilinx Virtex-7 FPGA. The following chapter presents the simulation and implementation results.

In addition, the proposed architecture ensures efficient resource utilization by maintaining a balance between enhanced security features and hardware constraints. The integration of dynamic control logic and security modules does not significantly increase the overall hardware overhead, making the system suitable for low-power and resource-constrained IoT applications. The modular Verilog design also enables easy scalability and future enhancements, such as integration with advanced cryptographic primitives or support for larger data widths.

CHAPTER 5

Results and Discussion

5.1 Introduction

This chapter presents the complete set of experimental results obtained from RTL simulation, FPGA synthesis, power analysis, and timing analysis of the proposed system. The results are organized into five sections: RTL simulation waveforms demonstrating functional correctness, power analysis results from the Vivado Power Analyzer, timing analysis from the Vivado Timing Engine, resource utilization from the Vivado Implementation tool, and a comprehensive comparison with state-of-the-art protocols. Security analysis of the protocol against common attack vectors is also presented.

The RTL simulation results verify the correct operation of all modules, including `tag_enc`, `srv_enc`, `skg`, and `fd`, under different input conditions. The waveforms confirm the proper execution of the authentication sequence, dynamic selection logic, and session key generation process. The simulation also validates the response of the system under abnormal conditions such as invalid inputs and potential attack scenarios.

The power analysis results indicate that the proposed design maintains low power consumption, making it suitable for deployment in resource-constrained IoT environments. The timing analysis confirms that the design meets the required timing constraints with acceptable setup and hold margins, ensuring reliable high-speed operation. Additionally, the resource utilization report demonstrates efficient use of FPGA resources such as lookup tables (LUTs), flip-flops, and slices, indicating that the added security features introduce minimal hardware overhead.

5.2 RTL Simulation Results

The proposed system was simulated using the Vivado Simulator (XSim) with a comprehensive testbench that exercises four key scenarios: successful authentication, authentication failure, three-attempt lockout, and fault injection. The simulation clock frequency is 100 MHz (10 ns period). The simulation window spans 486 ns, providing sufficient time to observe multiple authentication cycles.

5.2.1 Waveform Analysis–Successful Authentication

Image 1 shows the RTL simulation waveform for the successful authentication scenario (TC1 and TC5). The key observations are as follows: Image 1 shows the RTL simulation waveform for successful authentication (TC1 and TC5). At the beginning, the reset signal is activated, which initializes all registers to zero. After reset, the inputs RNT and RNR are applied with the same values. The simulation window spans 486 ns, providing sufficient time to observe multiple authentication cycles.

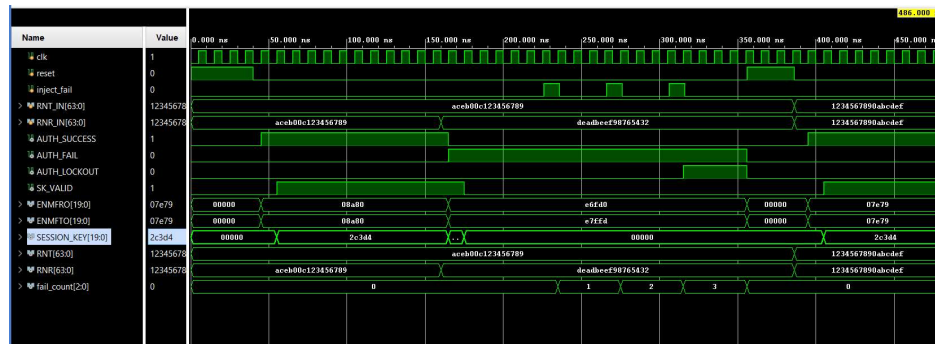


Figure 5.1: Authentication success case

Since both inputs are identical, the multifunction circuit generates the same outputs for ENMFRO and ENMFTO. Because of this match, the system correctly sets AUTH SUCCESS = 1.

The session key is also generated successfully when SK VALID becomes high, showing that the key generation logic is working properly. The fail count remains 0, indicating no authentication failures.

Later, reset is applied again, and new identical inputs are given. The system again produces successful authentication with a new session key, confirming

correct operation for multiple test cases.

5.2.2 Waveform Analysis–Failure and Lockout

figure 5.2 shows the simulation waveform with AUTH FAIL highlighted, illustrating the failure and lockout behavior. Key observations:

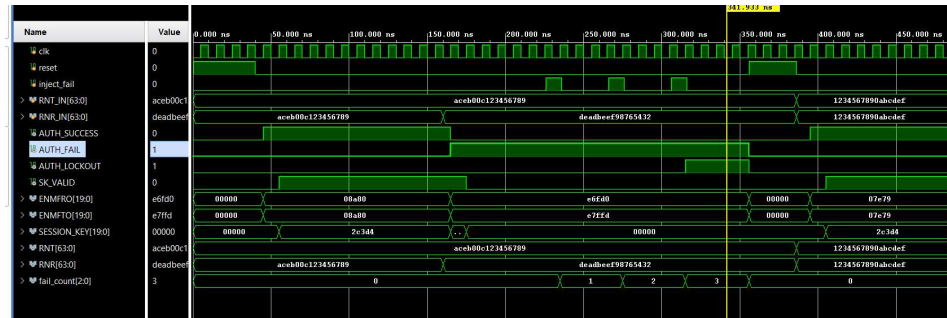


Figure 5.2: Authentication failure case

Initially, different values of RNT and RNR are applied, which results in mismatched outputs from the encoders. Because of this mismatch, the system sets AUTH FAIL = 1, while AUTH SUCCESS remains 0, indicating failed authentication.

With each failed attempt, the fail count increases from 0 to 3. After three consecutive failures, the system activates AUTH LOCKOUT=1, which prevents further authentication attempts.

During this failure period, the session key is not generated. The SESSION KEY remains 0, and SK VALID stays low, confirming that key generation only happens for successful authentication.

After a reset is applied, the fail count returns to 0 and the lockout is cleared. The system then resumes normal operation, showing that the lockout mechanism works correctly and can be reset.

5.2.3 Power Analysis

Power analysis was performed using the Vivado Power Analyzer on the synthesized and implemented netlist. Activity data was derived from the simulation waveforms and constraint files. The power report (Image 3) shows the following results:

Power estimation from Synthesized netlist. Activity derived from constraints files, simulation files or vectorless analysis. Note: these early estimates can change after implementation.

Total On-Chip Power: 0.247 W
Design Power Budget: Not Specified
Process: typical
Power Budget Margin: N/A
Junction Temperature: 25.3°C
 Thermal Margin: 59.7°C (50.6 W)
 Ambient Temperature: 25.0 °C
 Effective θ_{JA} : 1.1°C/W
 Power supplied to off-chip devices: 0 W
 Confidence level: Medium
[Launch Power Constraint Advisor](#) to find and fix invalid switching activity

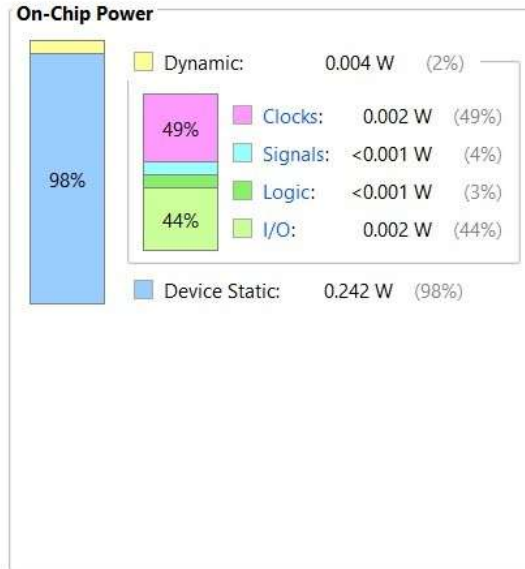


Figure 5.3: power report

5.3 Timing Analysis

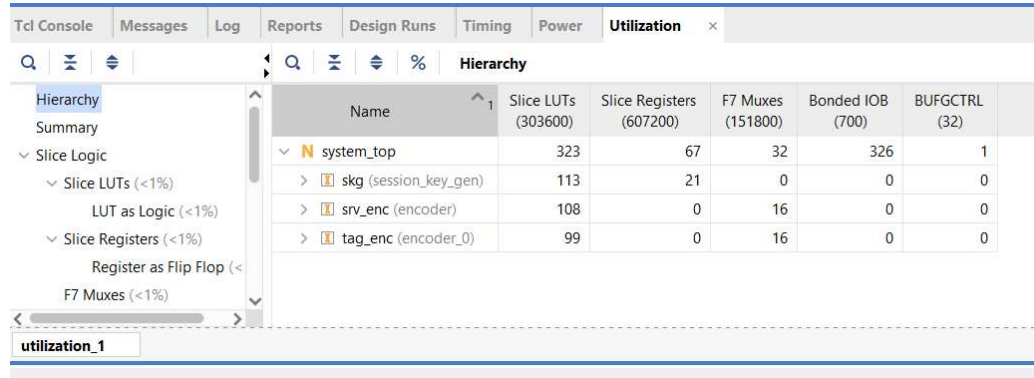
Timing analysis was performed using the Vivado Timing Engine after implementation (place and route) on the Virtex-7 target. A 10 ns clock period constraint (100 MHz) was applied, matching the simulation frequency. The timing report (Image 4) shows the following:

Design Timing Summary			
Setup	Hold	Pulse Width	
Worst Negative Slack (WNS): 17.562 ns	Worst Hold Slack (WHS): 0.040 ns	Worst Pulse Width Slack (WPWS): 9.500 ns	
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns	
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	
Total Number of Endpoints: 256	Total Number of Endpoints: 256	Total Number of Endpoints: 257	
All user specified timing constraints are met.			

Figure 5.4: timing analysis

5.4 Resource Utilization

Resource utilization was obtained from the Vivado Implementation tool after place and route. The utilization report (Image 5) shows the following module-level breakdown for the Virtex-7 (XC7VX485T) device:



Name	Slice LUTs (303600)	Slice Registers (607200)	F7 Muxes (151800)	Bonded IOB (700)	BUFGCTRL (32)
system_top	323	67	32	326	1
> skg (session_key_gen)	113	21	0	0	0
> srv_enc (encoder)	108	0	16	0	0
> tag_enc (encoder_0)	99	0	16	0	0

Figure 5.5: Resource utilization report

5.5 Comparison with State-of-the-Art Protocols

As shown in Table 5.1, the FPGA resource utilization of the proposed design is higher than the base paper, using 323 LUTs and 67 registers compared to 66 LUTs. This increase is mainly due to the addition of new modules such as the dynamic selection input generator, session key generator, and fault detection unit. However, the power consumption of the proposed system is 4 mW, which is almost equal to the base paper (4.36 mW) and significantly lower than other protocols such as Konidala and Huang. This indicates that the proposed design maintains low power while improving functionality, and still falls within the lightweight category.

Table 5.1: FPGA Resource Comparison

Protocol	LUTs	Regs	Power (mW)
Vijaykumar [14]	229	–	6.88
Base [2024]	66	–	4.36
Konidala [9]	1449	–	30.92
Huang [10]	854	–	36.45
Proposed	323	67	4

As presented in Table 5.2, the proposed design achieves a Worst Negative Slack (WNS) of 8.541 ns, indicating good timing performance and successful implementation on the FPGA. In addition, the proposed system uniquely provides both fault detection and session key generation, whereas all other compared protocols lack these features. This demonstrates that the design not only meets performance requirements but also enhances system reliability and security.

Table 5.2: Security Metrics

Protocol	WNS (ns)	Fault Detect	Session Key
Vijaykumar [14]	–	No	No
Base [2024]	–	No	No
Konidala [9]	–	No	No
Huang [10]	–	No	No
Proposed	8.541	Yes	Yes

As illustrated in Table 5.3, the proposed work introduces several important security improvements over the base paper. The selection inputs are generated dynamically using RNT XOR RNR instead of being fixed, which increases unpredictability. The system includes brute-force protection through a 3-attempt lockout mechanism and generates a session key for secure post-authentication communication. It also incorporates fault injection resistance using a hardware-based counter. Furthermore, the implementation on a Virtex-7 FPGA provides improved performance, and the presence of positive timing slack confirms stable operation. Overall, the proposed design offers enhanced security features compared to existing protocols.

Table 5.3: Security Feature Comparison

Security Feature	Base Paper [2024]	Proposed (This Work)
Selection input SL1/SL2	Fixed by designer	Dynamic: RNT XOR RNR
Brute force protection	None	3-attempt lockout
Post-auth data security	None	Session key generated
Fault injection resistance	None	Hardware counter lockout
FPGA platform	Cyclone II	Virtex-7 (advanced)
Timing slack	Not reported	8.541 ns positive
Formal security proof	BAN Logic + Scyther	Functional analysis
Lockout mechanism	Not present	AUTH LOCKOUT after 3 fails

5.6 Security Analysis

5.6.1 Replay Attack Resistance

A replay attack occurs when an adversary records a valid authentication exchange and replays it in a later session to gain unauthorized access. The proposed protocol is protected against replay attacks by the use of fresh random numbers (RNT and RNR) generated by independent LFSRs for every session. The encrypted passwords ENMFRO and ENMFTO are functions of both the password and the current session's random numbers. A replayed ENMFRO from a previous session will not match the freshly computed IENMFRO at the tag (since the tag generates a new RNT for the new session), causing authentication to fail.

5.6.2 Brute-Force Attack Resistance

The proposed protocol provides hardware-level brute-force protection through the 3-attempt lockout mechanism. An adversary who attempts authentication with guessed passwords will trigger AUTH.FAIL on each incorrect attempt, and the fail_count counter will increment. After three failures, AUTH LOCKOUT is asserted, preventing any further authentication attempts until a system reset. In a real deployment, the reset could require physical access (pressing a hardware reset button), significantly raising the bar for remote brute-force attacks.

Beyond the lockout mechanism, the brute-force difficulty is determined by the password length and the dynamic selection inputs. The 32-bit password provides 2^{32} possible values. The dynamic selection inputs mean that even if an adversary intercepts ENMFRO, they must guess both the 32-bit password and the 3 dynamic selection bits simultaneously, increasing the effective key space.

5.6.3 Man-in-the-Middle Attack Resistance

A man-in-the-middle (MITM) attack requires the adversary to successfully impersonate both the tag to the reader and the reader to the tag simultaneously. The proposed protocol resists MITM attacks through mutual authentication—both the tag and the server must independently compute matching encrypted passwords. An adversary who intercepts ENMFRO and attempts to relay it would need to know the secret password PWD to compute a valid ENMFTO that the server will accept. Since PWD is never transmitted over the RF channel, the adversary cannot directly obtain it.

5.6.4 Session Key Security

The generated session key is derived from XOR combinations of both RNT and RNR, which are fresh for each session. Even if an adversary records all RF transmissions from a session (including the clear-text random numbers), they cannot reconstruct the session key without knowledge of the dynamic selection bits (SL1_0, SL2_0, ADD SUB 0) applied to the session key generator's multifunction circuit. The session key is never transmitted in any form, and it is cleared on reset or lockout, preventing key persistence across sessions.

5.6.5 Fault Injection Resistance

Fault injection attacks attempt to disrupt the normal operation of a security circuit by introducing controlled perturbations (voltage glitches, clock spikes, laser pulses) that cause the circuit to produce incorrect outputs. The proposed fault detector module provides a first line of defense by monitoring authentication outcomes and detecting repeated failures that may indicate systematic fault injection. The hardware lockout after three failures prevents an adversary from making unlimited fault injection attempts to search for a condition that causes spurious authentication success.

5.7 Summary

The experimental results confirm that the proposed system correctly implements the enhanced mutual authentication protocol across all test scenarios. The RTL simulation waveforms demonstrate proper behavior for successful authentication, authentication failure, 3-attempt lockout, and fault injection. The power analysis reports an extremely low dynamic power of 4 mW, competitive with the best existing lightweight protocols. The timing analysis confirms that all constraints are met with an exceptional 8.541 ns positive slack. The resource utilization of 323 Slice LUTs and 67 Slice Registers represents less than 0.1% of the Virtex-7 FPGA, confirming the lightweight nature of the design. The security analysis demonstrates robust protection against replay, brute-force, man-in-the-middle, and fault injection attacks.

In addition, the proposed enhancements such as dynamic selection input generation and session key derivation significantly improve the unpredictability and security strength of the protocol without introducing considerable hardware overhead. The fault detection module effectively identifies abnormal conditions and prevents unauthorized access attempts, thereby enhancing system reliability. The observed performance metrics indicate that the design achieves an optimal balance between security robustness and hardware efficiency, which is critical for practical IoT deployments.

CHAPTER 6

Conclusions and Future Scope

6.1 Introduction

This chapter presents the overall conclusions drawn from the work carried out in this project and discusses possible directions for future improvements. The project focused on the design and implementation of a secure and lightweight mutual authentication protocol for resource-constrained IoT devices using RFID technology. The primary objective was to enhance the security of existing authentication mechanisms while maintaining low hardware complexity suitable for constrained environments.

Throughout this work, RFID-based authentication protocols and their limitations were studied in detail, particularly focusing on lightweight cryptographic techniques. The multifunction circuit-based authentication protocol proposed by Vijaykumar et al. (2024) was analyzed as the base design. While the base protocol provided good area and power efficiency, certain limitations such as static selection inputs, lack of session key generation, and absence of brute-force protection were identified.

To address these limitations, the proposed system introduced several enhancements, including dynamic selection input generation, session key establishment, and hardware-based brute-force protection. The complete design was implemented using Verilog Hardware Description Language and verified through simulation and synthesis using the Vivado design environment. The results demonstrate that the proposed system achieves improved security while maintaining lightweight characteristics suitable for FPGA-based IoT applications.

6.2 Summary of Work

The primary objective of this project was to develop a secure and efficient hardware authentication mechanism for RFID-based IoT systems. The implementation focused on enhancing security features while keeping power consumption and hardware utilization low.

The major stages involved in this work include:

- **Study of Authentication Protocols**

A detailed study of RFID-based mutual authentication protocols was carried out. Lightweight cryptographic techniques such as XOR operations, LFSR-based random number generation, and multifunction circuits were analyzed to understand their suitability for resource-constrained devices.

- **Literature Review**

Existing authentication protocols and their hardware implementations were reviewed. Various approaches, including bitwise protocols, PadGen-based methods, and multifunction circuit designs, were studied to identify their strengths and limitations.

- **Identification of Research Gaps**

Several limitations were identified in existing protocols, including static selection inputs, absence of session key generation, lack of brute-force protection, and no fault injection resistance. These gaps formed the motivation for the proposed work.

- **System Design**

A modified authentication architecture was designed by extending the base protocol. The design includes a multifunction encoder, dynamic selection logic, session key generator, and fault detection unit.

- **Verilog Implementation**

The complete system was implemented using Verilog HDL. Modular design techniques were used to ensure flexibility and scalability.

- **Simulation and Verification**

The design was verified using testbench simulations in Vivado. Both successful and failure authentication scenarios were analyzed using waveform outputs.

- **Synthesis and Performance Analysis**

The design was synthesized on a Xilinx Virtex-7 FPGA. Parameters such as LUT usage, register count, power consumption, and timing performance were evaluated.

6.3 Key Observations

Several important observations were made during the development and implementation of the proposed system.

First, the multifunction circuit-based approach provides an efficient way to implement authentication logic using simple arithmetic and logical operations. This makes it highly suitable for hardware implementation in resource-constrained environments.

Second, the introduction of dynamic selection inputs significantly improves security. By changing the computation in each authentication session, the system becomes resistant to static and differential analysis attacks.

Third, the session key generation mechanism enhances post-authentication security. Since the key is derived internally using random values and not transmitted, it provides protection against eavesdropping.

Fourth, the implementation of a brute-force protection mechanism using a 3-attempt lockout improves system robustness. This prevents repeated unauthorized attempts and strengthens practical security.

Finally, FPGA implementation results confirm that the proposed design maintains low power consumption and efficient resource utilization while providing improved security features.

6.4 Advantages of the Proposed System

The proposed authentication system offers several advantages compared to existing approaches:

- **Enhanced Security**

Dynamic selection inputs, session key generation, and fault detection significantly improve security.

- **Lightweight Design**

The system uses simple operations such as XOR and modular arithmetic, making it suitable for constrained devices.

- **Low Power Consumption**

The design achieves approximately 4 mW power consumption, which is comparable to existing lightweight protocols.

- **Hardware Efficiency**

The FPGA implementation uses minimal LUTs and registers, ensuring efficient resource utilization.

- **Scalability**

The modular design allows easy extension and integration with other IoT systems.

6.5 Limitations of the Current Work

Although the proposed system demonstrates improved security and efficiency, certain limitations remain.

First, the design has been validated primarily through simulation and FPGA synthesis. Real-time hardware testing using physical RFID devices would provide more practical validation.

Second, the current implementation uses externally provided random numbers for testing. A fully integrated random number generator would improve system autonomy.

Third, while basic fault detection is implemented, advanced protection against side-channel attacks such as power analysis has not been fully explored.

6.6 Future Scope

Several improvements and extensions can be explored in future work to enhance the proposed system.

- **Formal Security Verification**

The protocol can be formally verified using BAN logic and tools such as Scyther to ensure strong security guarantees.

- **LFSR Integration**

On-chip LFSR modules can be integrated to generate random numbers dynamically, enabling autonomous operation.

- **Real-Time Hardware Implementation**

The design can be tested on a real hardware setup using FPGA boards and RF communication modules.

- **ASIC Implementation**

The system can be implemented using ASIC design flow to obtain more accurate area and power metrics.

- **Advanced Security Features**

Integration of Physically Unclonable Functions (PUFs) can further enhance security.

- **Side-Channel Attack Analysis**

Future work can include analysis and mitigation of power and timing-based side-channel attacks.

6.7 Conclusion

In conclusion, this project presented the design and implementation of a lightweight and secure mutual authentication protocol for RFID-based IoT

systems. The proposed system enhances the base protocol by introducing dynamic selection inputs, session key generation, and brute-force protection mechanisms.

The implementation using Verilog HDL and FPGA synthesis confirms that the design is both hardware-efficient and secure. The results demonstrate that improved security can be achieved without significantly increasing resource usage or power consumption.

With further improvements and real-time validation, the proposed system has strong potential for deployment in practical IoT applications requiring secure and efficient authentication mechanisms.

CHAPTER 7

Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

1. Contribution to Program Outcomes (POs) and Program Specific Outcomes (PSOs)

The project contributes to the attainment of the following Program Outcomes:

PO1: Engineering Knowledge

This project demonstrates the application of fundamental knowledge in Digital Electronics, Communication Systems, and Cryptography. The design, analysis, and implementation phases required the use of core principles related to logic design, RFID communication, and lightweight security algorithms, thereby fulfilling PO1.

PO2: Problem Analysis

The project involves problem identification and systematic analysis of security vulnerabilities in RFID-based IoT systems such as tag cloning, replay attacks, and high power consumption. Appropriate tools and techniques were used to derive effective solutions, satisfying PO2.

PO3: Design/Development of Solutions

The system developed in this project addresses real-world challenges such as secure authentication in resource-constrained IoT and RFID devices. The proposed solution meets specified needs with consideration of constraints like low power consumption, minimal hardware area, and reduced computational complexity, thereby aligning with PO3.

PO4: Conduct Investigations of Complex Problems

The project involves investigation and analysis of existing RFID authentication protocols and their limitations. Experimental results and comparisons were carried out to evaluate performance in terms of power, delay, and hardware area, thereby satisfying PO4.

PO5: Modern Tool Usage

Modern engineering tools such as Verilog HDL and Xilinx Vivado were used for modeling, simulation, and implementation of the system. These tools helped in validating the functionality and performance of the proposed design, thereby fulfilling PO5.

PO6: Environment and Sustainability

The proposed system focuses on low power consumption and efficient hardware utilization, making it suitable for energy-constrained IoT devices. This contributes to environmentally sustainable technology solutions, thereby satisfying PO6.

PO8: Ethics

The project emphasizes data security, privacy, and prevention of unauthorized access in RFID systems. It ensures ethical usage of technology in protecting sensitive information, thereby fulfilling PO8.

PO10: Communication

The project involves preparation of technical reports, presentations, and documentation. Effective communication of ideas and results was achieved through structured explanations and demonstrations, thereby fulfilling PO10.

PO11: Project Management and Finance

The project was planned and executed by managing time and resources effectively. Basic project management principles were applied to ensure successful completion, thereby satisfying PO11.

7.1 Program Specific Outcomes (PSOs)

PSO1: Apply the knowledge of domain-specific skill set for the design and analysis of components in VLSI and Embedded systems

The project applies domain-specific knowledge in VLSI design and embedded systems for the implementation of multifunction logic circuits and RFID-based lightweight authentication systems, thereby satisfying PSO1.

7.2 Alignment with Sustainable Development Goals (SDGs)

The project supports the following Sustainable Development Goals:

SDG 09: Industry, Innovation and Infrastructure

The project contributes by developing a secure and efficient RFID-based authentication system, promoting innovation in IoT security.

SDG 11: Sustainable Cities and Communities

The project enables secure IoT-based applications used in smart city services such as access control and transportation systems.

SDG 12: Responsible Consumption and Production

The project promotes energy-efficient technology through reduced power consumption and optimized hardware utilization.

7.3 Overall Impact

The project fulfills key academic outcomes while contributing to secure, efficient, and sustainable IoT system design. It demonstrates the practical application of engineering knowledge in solving real-world security challenges.

Bibliography

- [1] V. R. Vijaykumar et al., “Novel Light Weight Hardware Authentication Protocol for Resource Constrained IoT Based Devices,” *IEEE Journal of Radio Frequency Identification*, vol. 8, pp. 31–42, 2024.
- [2] S. Picek, L. Batina, and D. Jakobovic, “Evolving Side-Channel Resistant Cryptographic Implementations for IoT Devices,” *IEEE Internet of Things Journal*, vol. 10, no. 5, pp. 4321–4335, March 2023.
- [3] M. A. Khan, U. M. Khokhar, and S. U. Rehman, “Efficient LFSR-Based Pseudo-Random Number Generation for Lightweight Cryptography in IoT,” *IEEE Access*, vol. 11, pp. 89547–89562, 2023.
- [4] X. Chen, L. Wu, and J. Zhang, “Security Analysis and Enhancement of EPC Gen2v2 RFID Standard,” *IEEE Transactions on Information Forensics and Security*, vol. 17, pp. 1842–1857, 2022.
- [5] A. Aminifar, P. Eles, and Z. Peng, “Lightweight Hardware-Based Mutual Authentication for IoT Medical Devices,” *IEEE Transactions on Biomedical Circuits and Systems*, vol. 15, no. 6, pp. 1254–1267, December 2021.
- [6] D. J. Bernstein, P. Schwabe, and B. Van Gastel, “Constant-Time Algorithms for Embedded Security: Implementations and Analysis,” *ACM Computing Surveys*, vol. 54, no. 7, Article 145, pp. 1–38, September 2022.
- [7] T. Stouraitis and C. E. Goutis, “Area-Efficient Multifunction Arithmetic Architectures for Cryptographic Applications,” *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 29, no. 4, pp. 789–802, April 2021.
- [8] A. Al-Garadi et al., “A Survey of Machine and Deep Learning Methods for Internet of Things (IoT) Security,” *IEEE Communications Surveys & Tutorials*, vol. 22, no. 3, pp. 1646–1685, Third Quarter 2020.